

Digestion and Absorption of Carbohydrates

From the Mouth to the Stomach

The mechanical and chemical digestion of carbohydrates begins in the mouth. Chewing, also known as mastication, crumbles the carbohydrate foods into smaller and smaller pieces. The salivary glands in the oral cavity secrete saliva that coats the food particles. Saliva contains the enzyme salivary amylase. This enzyme breaks the bonds between the monomeric sugar units of disaccharides, oligosaccharides, and starches. The salivary amylase breaks down amylose and amylopectin into smaller chains of glucose called dextrins and maltose. The increased concentration of maltose in the mouth that results from the mechanical and chemical breakdown of starches in whole grains is what enhances their sweetness. Only about five percent of starches are broken down in the mouth (which is a good thing as more glucose in the mouth would lead to more tooth decay). When carbohydrates reach the stomach, no further chemical breakdown occurs because the amylase enzyme does not function in the acidic conditions of the stomach. However, mechanical breakdown is ongoing—the strong peristaltic contractions of the stomach mix the carbohydrates into a more uniform mixture of chyme.

Salivary glands secrete salivary amylase, which begins the chemical breakdown of carbohydrates by breaking the bonds between monomeric sugar units.

From the Stomach to the Small Intestine

The chyme is gradually expelled into the upper part of the small intestine, where most carbohydrate digestion occurs. Upon entry of the chyme into the small intestine, the pancreas releases pancreatic juice through the pancreatic duct. This pancreatic juice contains the enzyme pancreatic amylase, which starts again the breakdown of dextrins and any remaining polysaccharides into shorter and shorter carbohydrate chains (e.g., maltose). Additionally, the enzymes sucrase, maltase, and lactase are secreted by the intestinal cells that line the villi. Sucrase breaks

sucrose into glucose and fructose molecules. Maltase breaks the bond between the two glucose units of maltose, and lactase breaks the bond between galactose and glucose. Once carbohydrates are chemically broken down into single sugar units, they are then transported to the inside of intestinal cells and into the bloodstream.

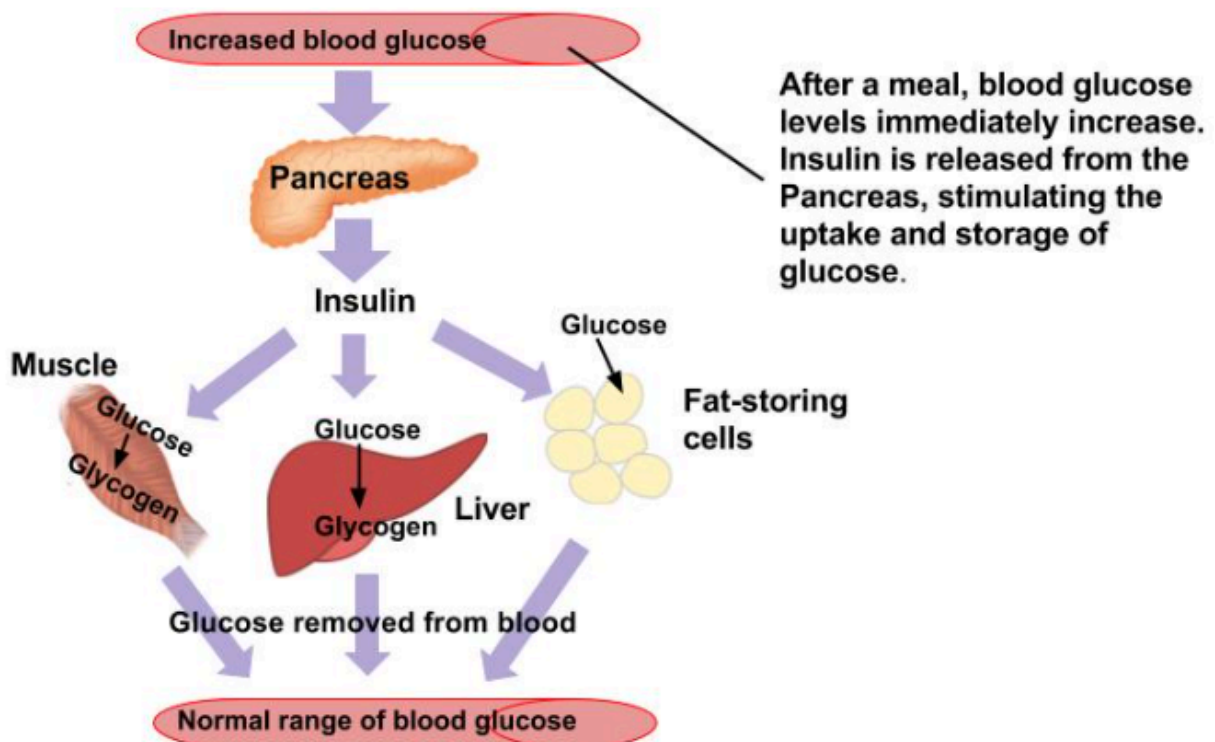
Absorption: Going to the Blood Stream

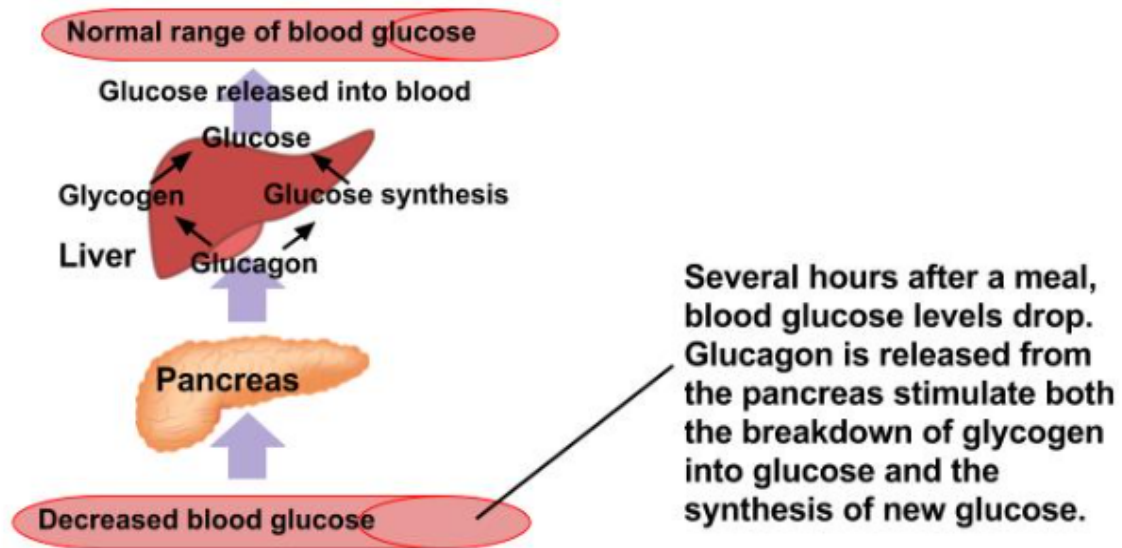
The cells in the small intestine have membranes that contain many transport proteins to get the monosaccharides and other nutrients into the blood where they can be distributed to the rest of the body. The first organ to receive glucose, fructose, and galactose is the liver. The liver takes them up and converts galactose to glucose, breaks fructose into even smaller carbon-containing units, and either stores glucose as glycogen or exports it back to the blood. How much glucose the liver exports to the blood is under hormonal control and you will soon discover that even the glucose itself regulates its concentrations in the blood. If we don't immediately need glucose for energy, it is stored in the form of glycogen in the muscle (approx. 300 g) and liver (approx. 100 g). Liver stores serve to top up blood glucose through glycogen breakdown (glycogenolysis) and muscle glycogen is mainly accessed during exercise. If glycogen levels drop, our bodies undergo gluconeogenesis (the generation of glucose from a non-carbohydrate source) to generate glucose.

Maintaining Blood Glucose Levels: The Pancreas and Liver

Glucose levels in the blood are tightly controlled, as having either too much or too little glucose in the blood can have negative health consequences. Glucose regulates its levels in the blood via a process called negative feedback. An everyday example of negative feedback is in your oven because it contains a thermostat. When you set the temperature to cook a delicious homemade noodle casserole at 375°F, the thermostat senses the temperature and sends an electrical signal to turn the elements on and heat up the oven. When the temperature reaches 375°F, the thermostat senses the temperature and sends a signal to turn the element off. Similarly, your body senses blood glucose levels and maintains the glucose “temperature” in the target range. The glucose thermostat is located within the cells of the pancreas. After eating a meal containing carbohydrates, glucose levels rise in the blood.

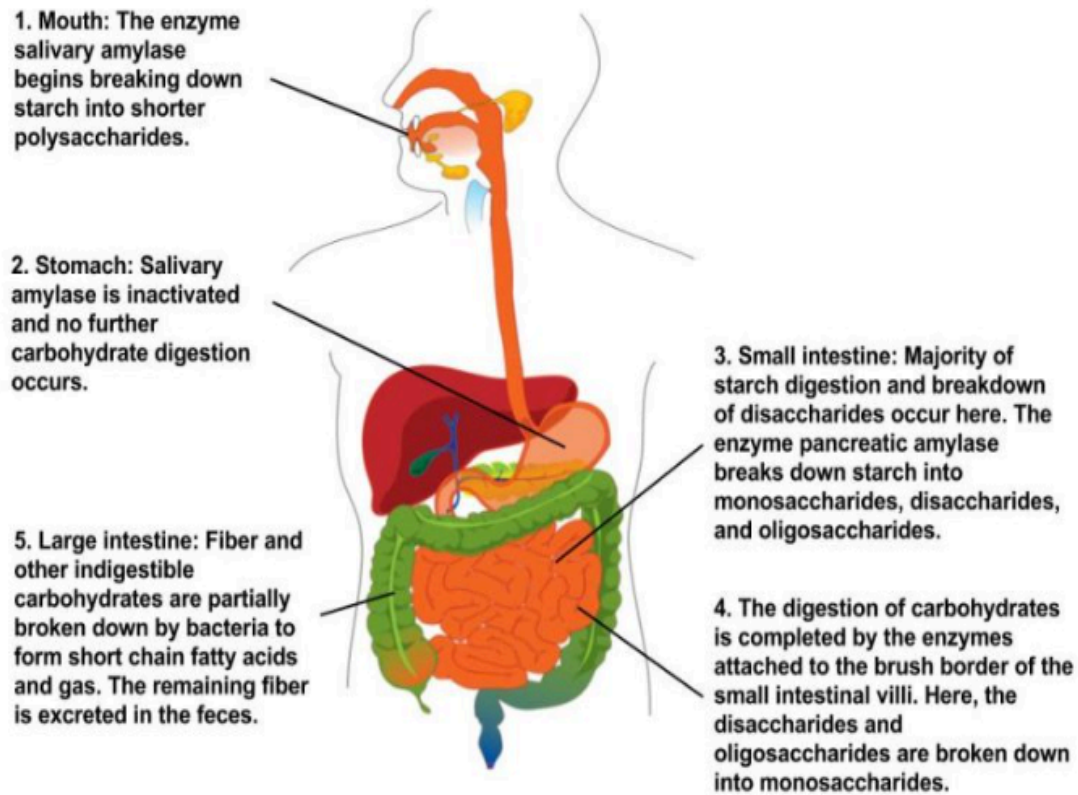
Insulin-secreting cells in the pancreas sense the increase in blood glucose and release the hormone insulin into the blood. Insulin sends a signal to the body's cells to remove glucose from the blood by transporting it into different organ cells around the body and using it to make energy. In the case of muscle tissue and the liver, insulin sends the biological message to store glucose away as glycogen. The presence of insulin in the blood signifies to the body that glucose is available for fuel. As glucose is transported into the cells around the body, the blood glucose levels decrease. Insulin has an opposing hormone called glucagon. Glucagon-secreting cells in the pancreas sense the drop in glucose and, in response, release glucagon into the blood. Glucagon communicates to the cells in the body to stop using all the glucose. More specifically, it signals the liver to break down glycogen and release the stored glucose into the blood, so that glucose levels stay within the target range and all cells get the needed fuel to function properly. Glucagon also stimulates gluconeogenesis. Catecholamines, such as epinephrine and norepinephrine, as well as cortisol and growth hormone, also work to increase blood glucose.





Leftover Carbohydrates: The Large Intestine

Almost all of the carbohydrates, except for dietary fibre and resistant starches, are efficiently digested and absorbed into the body. Some of the remaining indigestible carbohydrates are broken down by enzymes released by bacteria in the large intestine. The products of bacterial digestion of these slow-releasing carbohydrates are short-chain fatty acids and some gases. The short-chain fatty acids are either used by the bacteria to make energy and grow, are eliminated in the feces, or are absorbed into cells of the colon, with a small amount being transported to the liver. Colonic cells use short-chain fatty acids to support some of their functions. The liver can also metabolize short-chain fatty acids into cellular energy. Dietary fibre yields about 2 kilocalories per gram of energy for humans but is highly dependent upon the fibre type, with soluble fibres and resistant starches yielding more energy than insoluble fibres. Since dietary fibre is digested much less in the gastrointestinal tract than other carbohydrate types (simple sugars, many starches), the rise in blood glucose after eating them is less and slower. These physiological attributes of high-fibre foods (i.e. whole grains) are linked to a decrease in weight gain and reduced risk of chronic diseases, such as Type 2 diabetes and cardiovascular disease.

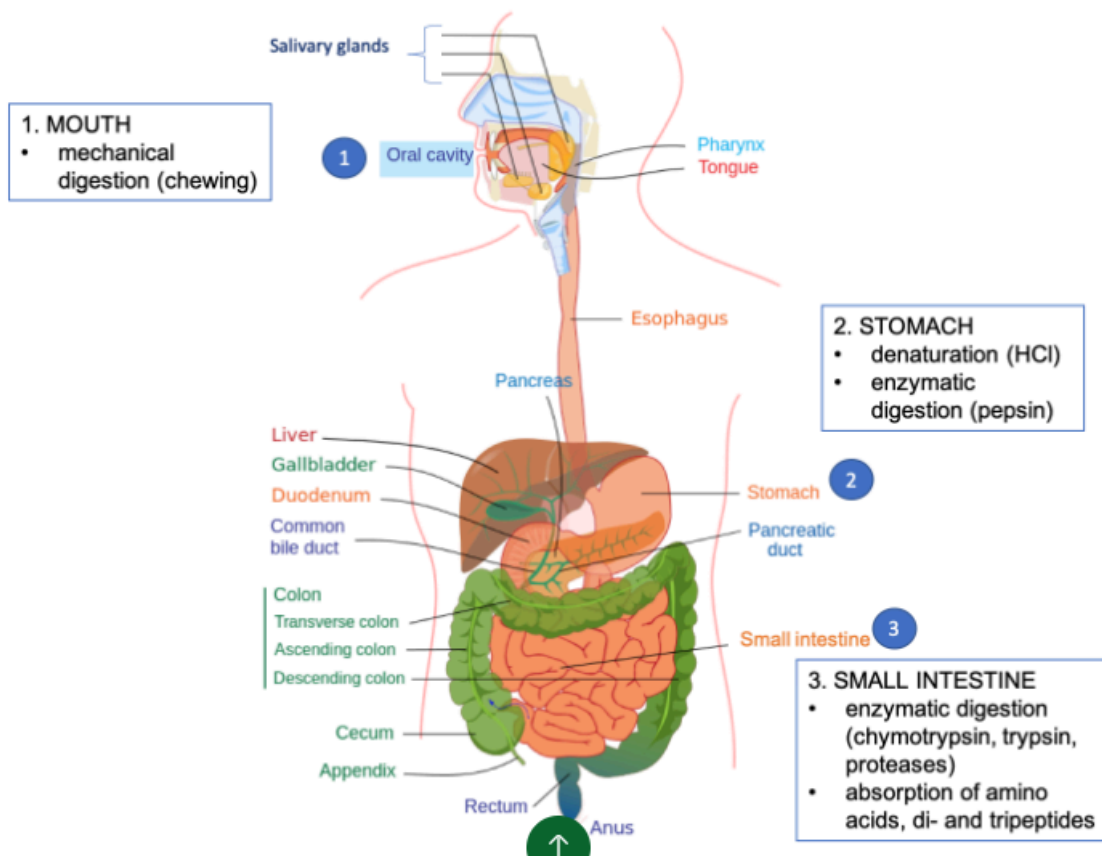


Protein Digestion and Absorption

When you eat food, the body's digestive system breaks down dietary protein into individual amino acids, which are absorbed and used by cells to build other proteins and a few other macromolecules, such as DNA. Let's follow the path that proteins take down the gastrointestinal tract and into the circulatory system.

Eggs are a good dietary source of protein and will be used as our example as we discuss the processes of digestion and absorption of protein. One egg, whether raw, hard-boiled, scrambled, or fried, supplies about six grams of protein.

In the image below, follow the numbers to see what happens to the protein in our egg at each site of digestion.



1 – Protein digestion in the mouth

Unless you are eating it raw, the first step in digesting an egg (or any other solid food) is chewing. The teeth begin the **mechanical breakdown** of large egg pieces into smaller pieces that can be swallowed. The salivary glands secrete saliva to aid swallowing and the passage of the partially mashed egg through the esophagus.

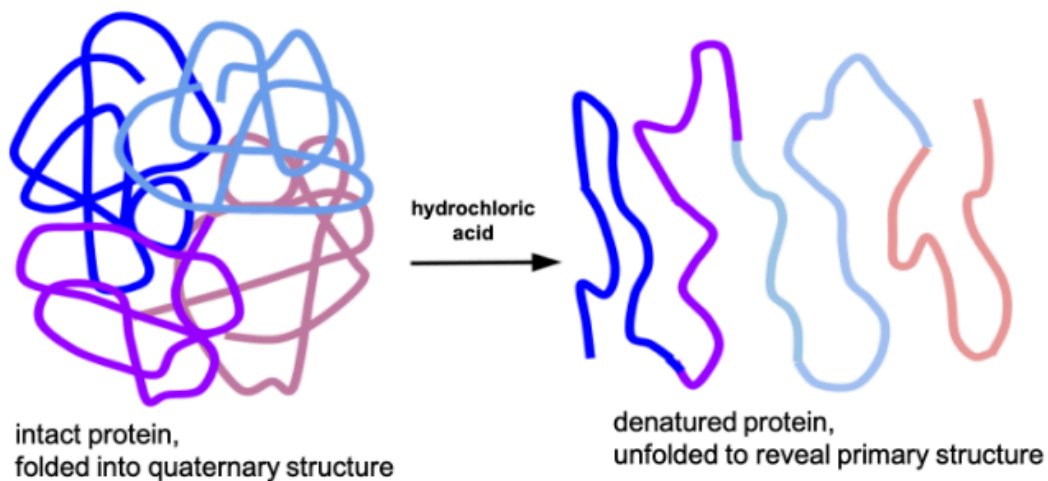
2 – Protein digestion in the stomach

The mashed egg pieces enter the stomach from the esophagus. As illustrated in the image below, both mechanical and chemical digestion take place in the stomach. The stomach releases gastric juices containing **hydrochloric acid** and the enzyme, **pepsin**, which initiate the chemical digestion of protein. Muscular contractions, called peristalsis, also aid in digestion. The powerful stomach

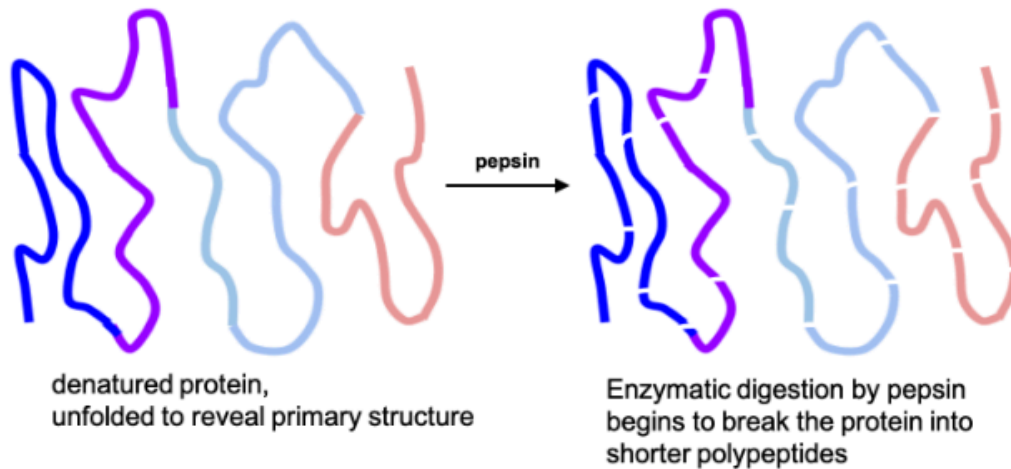
contractions churn the partially digested protein into a more uniform mixture, which is called chyme.



Because of the hydrochloric acid in the stomach, it has a very low pH of 1.5-3.5. **The acidity of the stomach causes food proteins to denature, unfolding their three-dimensional structure** to reveal just the polypeptide chain. This is the first step of chemical digestion of proteins. Recall that the three-dimensional structure of a protein is essential to its function, so **denaturation in the stomach also destroys protein function**. (This is why a protein such as insulin can't be taken as an oral medication. Its function is destroyed in the digestive tract, first by denaturation and then further by enzymatic digestion. Instead, it has to be injected so that it is absorbed intact into the bloodstream.)



Once proteins are denatured in the stomach, the peptide bonds linking amino acids together are more accessible for enzymatic digestion. That process is started by *pepsin*, an enzyme that is secreted by the cells that line the stomach and is activated by hydrochloric acid. Pepsin begins breaking peptide bonds, creating shorter polypeptides.



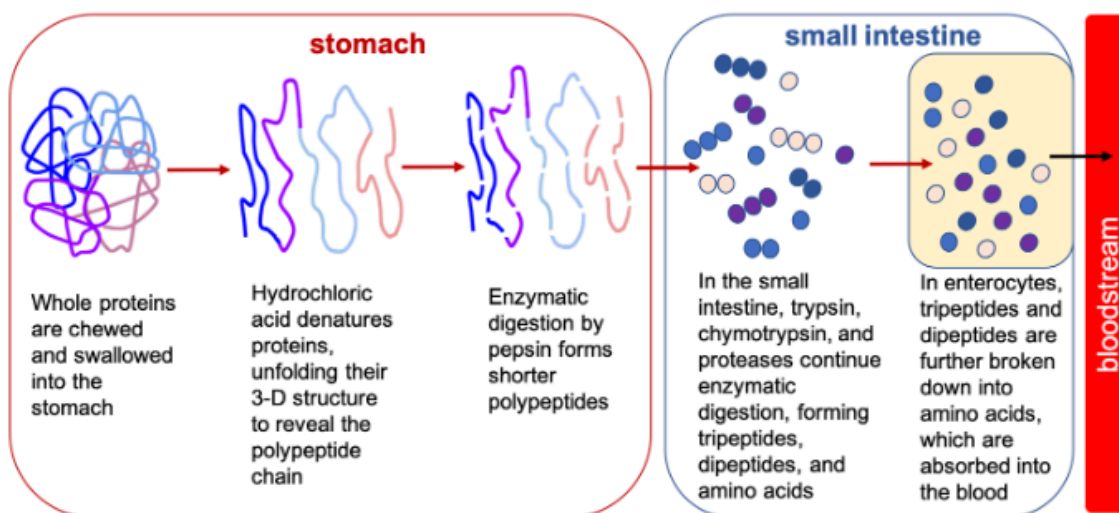
Proteins are large globular molecules, and their chemical breakdown requires time and mixing. Protein digestion in the stomach takes a longer time than carbohydrate digestion, but a shorter time than fat digestion. Eating a high-protein meal increases the amount of time required to sufficiently break down the meal in the stomach. Food remains in the stomach longer, making you feel full longer.

3 – Protein digestion and absorption in the small intestine

The chyme leaves the stomach and enters the small intestine, where the majority of protein digestion occurs. The pancreas secretes digestive juices into the small intestine, and these contain more enzymes to further break down polypeptides.

The two major pancreatic enzymes that digest proteins in the small intestine are *chymotrypsin* and *trypsin*. Trypsin activates other protein-digesting enzymes called *proteases*, and together, **these enzymes break proteins down to tripeptides, dipeptides, and individual amino acids.** The cells that line the small intestine release additional enzymes that also contribute to the enzymatic digestion of polypeptides.

Tripeptides, dipeptides, and single amino acids enter the enterocytes of the small intestine using active transport systems, which require ATP. **Once inside, the tripeptides and dipeptides are all broken down to single amino acids, which are absorbed into the bloodstream.** There are several different types of transport systems to accommodate different types of amino acids. Amino acids with structural similarities end up competing to use these transporters. That's not a problem if your protein is coming from food, because it naturally contains a mix of amino acids. However, if you take high doses of amino acid supplements, those could theoretically interfere with absorption of other amino acids.



Proteins that aren't fully digested in the small intestine pass into the large intestine and are eventually excreted in the feces. Recall from the last page that plant-based proteins are a bit less digestible than animal proteins, because some proteins are bound in plant cell walls.

What happens to absorbed amino acids?

Once the amino acids are in the blood, they are transported to the liver. As with other macronutrients, the liver is the checkpoint for amino acid distribution and any further breakdown of amino acids, which is very minimal. Dietary amino acids then become part of the body's amino acid pool.

Assuming the body has enough glucose and other sources of energy, those amino acids will be used in one of the following ways:

- Protein synthesis in cells around the body
- Making nonessential amino acids needed for protein synthesis
- Making other nitrogen-containing compounds
- Rearranged and stored as fat (there is no storage form of protein)

If there is not enough glucose or energy available, amino acids can also be used in one of these ways:

- Rearranged into glucose for fuel for the brain and red blood cells
- Metabolized as fuel, for an immediate source of ATP

In order to use amino acids to make ATP, glucose, or fat, the nitrogen first has to be removed in a process called *deamination*, which occurs in the liver and kidneys. The nitrogen is initially released as ammonia, and because ammonia is toxic, the liver transforms it into urea. Urea is then transported to the kidneys and excreted in the urine. Urea is a molecule that contains two nitrogens and is highly soluble in water. This makes it ideal for transporting excess nitrogen out of the body.

Because amino acids are building blocks that the body reserves in order to synthesize other proteins, more than 90 percent of the protein ingested does not get broken down further than the amino acid monomers.